Module II: Styling Pages

Study Notes · MAT5FS112 · Syllabus units 5–7

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1. Fonts and text styling

Font commands change the look of a piece of text:

COMMAND	RESULT
`\textbf{bold}`	**bold**
`\textit{italic}`	*italic*
`\emph{emphasis}`	*emphasis* (italic inside italic text becomes upright)
`\texttt{typewriter}`	`typewriter`
`\textsf{sans serif}`	sans serif
`\textsc{small caps}`	SMALL CAPS
`\underline{underlined}`	underlined

Use `\emph` for meaning and `\textit` for pure styling. Commands can be combined: `\textbf{\textit{both}}`, which gives ***both***.

1.1 Font size

Sizes are set with declarations. A declaration applies until the end of the enclosing group (a pair of braces or an environment):

```
{\small This is small text.}
{\Large This is large text.}
```

Sizes, from small to large: `\tiny`, `\scriptsize`, `\footnotesize`, `\small`, `\normalsize`, `\large`, `\Large`, `\LARGE`, `\huge`, `\Huge`.

This is small text. This is large text.

2. Spacing and page layout

2.1 Paragraph spacing

```
\setlength{\parindent}{0pt}     % no indent at the start of paragraphs
\setlength{\parskip}{0.6em}     % vertical gap between paragraphs
\noindent                       % suppress indent for one paragraph
```

2.2 Manual spacing

`\vspace{1cm}` adds vertical space and `\hspace{2cm}` adds horizontal space. `\\` forces a line break and `\newpage` forces a page break.

2.3 Margins

The `geometry` package sets margins in one line, as in the source file of these notes:

```
\usepackage[margin=2.5cm]{geometry}
```

2.4 Headers, footers and page numbers

The `fancyhdr` package gives control over the running header and footer, with `L`, `C` and `R` positions. These notes use:

```
\pagestyle{fancy}
\fancyhf{}                          % clear everything first
\fancyhead[L]{Left text}
\fancyhead[R]{\thepage}
\renewcommand{\headrulewidth}{0.4pt}
```

A preview of the resulting running header:

Page numbering style: `\pagenumbering{roman}` (i, ii, iii…) or `\pagenumbering{arabic}`. `\thispagestyle{empty}` removes the header from one page.

3. Displayed text

Environments display blocks of text with a specific alignment or style.

3.1 Alignment

`center`
Centred text.

`flushleft`
Left-aligned (ragged right) text.

`flushright`
Right-aligned text.

3.2 Quotations and verse

The `quote` environment sets a paragraph with extra margin on both sides:

> *A quotation: a paragraph set with extra margin on both sides.*

The `quotation` environment is similar, but it indents the first line of each paragraph inside it.

`verse` keeps each line break you type and centres nothing else on the page:

> *Verse keeps each line break you type*
> *and centres nothing else on the page.*

3.3 Side-by-side blocks with minipage

Two `minipage` boxes of fixed width, placed one after the other with `\hfill` between them, sit side by side:

Left column. A `minipage` is a box of fixed width that holds text. Its contents are wrapped inside the box.

Right column. Two minipages placed one after the other with `\hfill` between them sit side by side.

3.4 Verbatim text

```
This text is printed exactly as typed:  $x^2$ and \textbf{stay raw}.
```

4. Rows and columns: tabular

The `tabular` environment sets rows and columns. The argument gives one letter per column: `l` (left), `c` (centre), `r` (right), and `p{width}` (wrapped paragraph of given width). Cells are separated by `&` and rows end with `\\`.

```
\begin{tabular}{lcr}
Item   & Mark & Grade \\
\hline
Algebra & 18  & A     \\
Real    & 14  & B     \\
\end{tabular}
```

Rendered:

ITEM	MARK	GRADE
Algebra	18	A
Real	14	B

4.1 Rules: `\hline` and `\cline`

`\hline` draws a full-width horizontal line. `\cline{2-3}` draws a line only under columns 2 to 3. Vertical lines come from `|` in the column specification.

```
\begin{tabular}{|l|c|c|}
\hline
Name & Physics & Maths \\
\cline{2-3}
Asha & 80      & 75    \\
\hline
\end{tabular}
```

Rendered:

NAME	PHYSICS	MATHS
Asha	80	75

4.2 Spanning columns: `\multicolumn`

```
\begin{tabular}{|l|c|c|}
\hline
\multicolumn{3}{|c|}{Semester V marks} \\
\hline
Name & Physics & Maths \\
\hline
Asha & 80      & 75    \\
\hline
\end{tabular}
```

Rendered:

SEMESTER V MARKS		
NAME	PHYSICS	MATHS
Asha	80	75

4.3 Professional tables with `booktabs`

The `booktabs` package uses `\toprule`, `\midrule`, `\bottomrule` and avoids vertical lines, which gives cleaner results. `\cmidrule{2-3}` draws a partial rule.

```
\begin{tabular}{l r r}
\toprule
Module & Units & Min.\ marks \\
\midrule
I   & 4 & 10 \\
II  & 3 & 10 \\
III & 4 & 10 \\
\bottomrule
\end{tabular}
```

Rendered:

MODULE	UNITS	MIN. MARKS
I	4	10
II	3	10
III	4	10

4.4 Fixed-width columns with `p{}`

Use `p{width}` when a cell has a paragraph of text. The `array` package adds `m{width}` (vertically centred) and `b{width}` (bottom aligned).

```
\begin{tabular}{p{3.5cm} p{8cm}}
\toprule
Command & Purpose \\
\midrule
\verb|\hline| & Draws a full horizontal rule across the table. \\
\verb|\multicolumn| & Merges several cells in a row into one. \\
\bottomrule
\end{tabular}
```

Rendered:

COMMAND	PURPOSE
`\hline`	Draws a full horizontal rule across the table.
`\multicolumn`	Merges several cells in a row into one.

5. Tables as floating objects

A table placed in a `table` environment floats: LaTeX moves it to a suitable place and numbers it. The caption and the label let the text refer to it.

```
\begin{table}[htbp]
  \centering
  \caption{Marks awarded in the LaTeX internal examination}
  \label{tab:marks}
  \begin{tabular}{lcc}
    \toprule
    Student & Assignment & Viva \\
    \midrule
    Student A & 9 & 8 \\
    Student B & 7 & 9 \\
    \bottomrule
  \end{tabular}
\end{table}
```

Rendered:

STUDENT	ASSIGNMENT	VIVA
Student A	9	8
Student B	7	9

Table 1: Marks awarded in the LaTeX internal examination

Placement. As `Table~\ref{tab:marks}` shows, the marks are recorded in two parts. The placement specifier `[htbp]` means: here, top, bottom, or on its own page. The `~` before `\ref` is a non-breaking space that keeps “Table” and the number together.